

5G GMLC Description

NLG

User Guide

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1 Introduction

The Gateway Mobile Location Center (GMLC) deployed in 5G core (5GC) network supports control plane positioning for subscribers in the standalone New Radio (NR) network.

2 Technical Description

2.1 GMLC Overview

The NLG implements the GMLC Network Function (NF) defined by 3GPP standards. This section gives an overview of the 3GPP Location Services (LCS) architecture and the interfaces related to the GMLC.

The 5GC architecture is service-based. The system functionality in Service-Based Architecture (SBA) is achieved with a set of NFs that provide service access to other authorized NFs. Authorization can be achieved by enabling OAuth 2.0 support. An NF is a component of the network infrastructure that provides a specific functional behavior, such as location determination. An NF is made up of NF services.

The interactions between NFs are represented in the following ways:

- Reference point representation
- Service-based representation

Figure 1 shows the reference point representation of 5G LCS defined by 3GPP TS 23.501, where the interaction between NFs is described by point-to-point reference points.

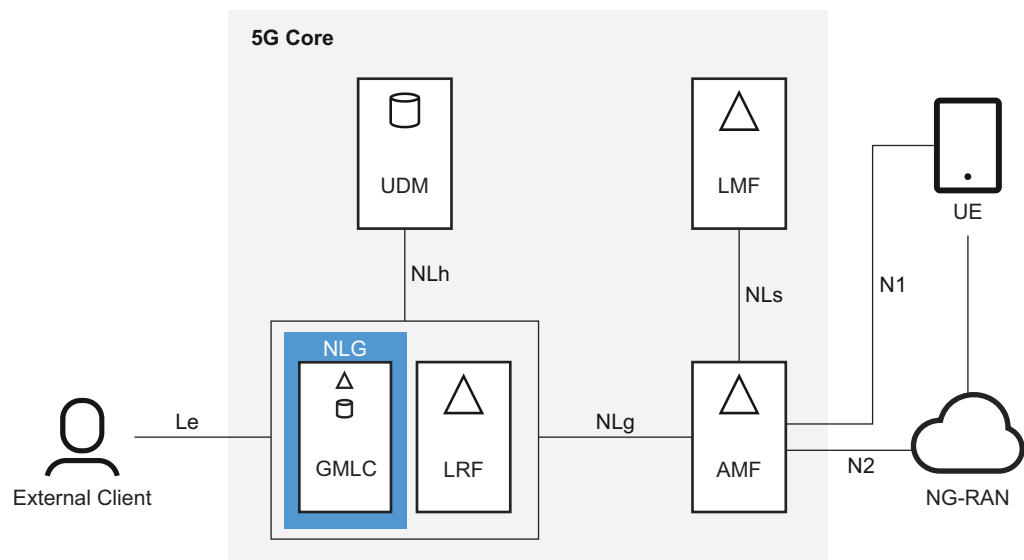


Figure 1 5G LCS Reference Point Representation

The following reference points support LCS in 5GC:



- N1: Reference point between a User Equipment (UE) and the Access and Mobility Management Function (AMF) through the Non-Access-Stratum (NAS)
- N2: Reference point between the New Generation - Radio Access Network (NG-RAN) and the AMF
- Le: Reference point between a GMLC and an LCS client

The following reference points are realized by Service-Based Interfaces (SBI):

- NLg: Reference point between a GMLC and the AMF
- NLs: Reference point between the AMF and the Location Management Function (LMF)
- NLh: Reference point between a GMLC and the User Data Management (UDM)

Figure 2 shows the service-based representation of the 5G LCS architecture, where the network functions are interconnected through SBIs.

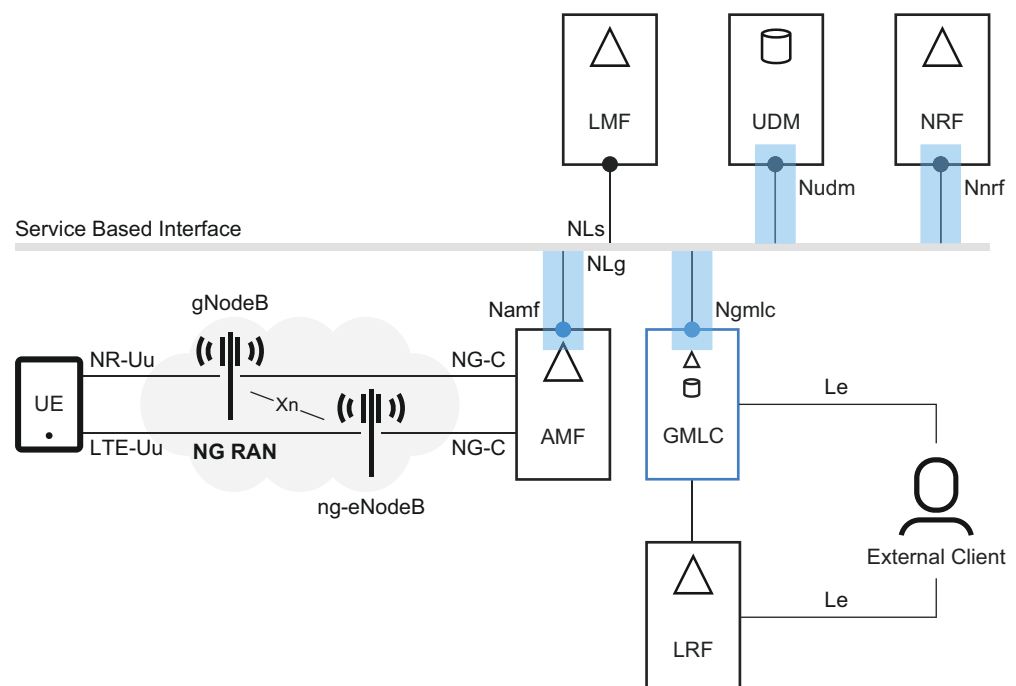


Figure 2 Service-based Representation of 5G LCS Architecture

The following SBIs are available for LCS:

- N1mf: SBI exhibited by LMF

- `Namf`: SBI exhibited by AMF
- `Nudm`: SBI exhibited by UDM
- `Nnrf`: SBI exhibited by Network Repository Function (NRF)

The 5G GMLC enables the external client to request the location of a target UE in the 5G network. The 5G GMLC uses the `Nudm` interface provided by the UDM to request information about the AMF node that is connected to the given UE.

The 5G GMLC also uses the `Namf` interface provided by the AMF to request positioning information of a specific UE. The 5G GMLC also interacts with the NRF through the `Nnrf` interface for NF service framework functions.

The 5G GMLC can also receive NI-LR notifications on the `Namf` interface about positioning results performed at emergency call setup and send the location information to the LCS client.

The 5G GMLC supports sending user-agent header on the `Namf`, the `Nnrf`, and the `Nudm` interfaces with a configurable value. For more information, see the *Http2* section in *Parameter List* for NLG.

According to the *3GPP 23.501* standard communication models, there can be direct or indirect communication performed with external NFs. In the latter case, the communication is done through the Service Communication Proxy (SCP). GMLC is prepared to communicate with AMF and UDM indirectly.

GMLC only supports indirect communication model C.

GMLC does not support the following:

- indirect communication model D
- the delegation of NF producer selection to the SCP

[Figure 3](#) shows the two cases of the SCP being deployed between the NF consumer and producer.

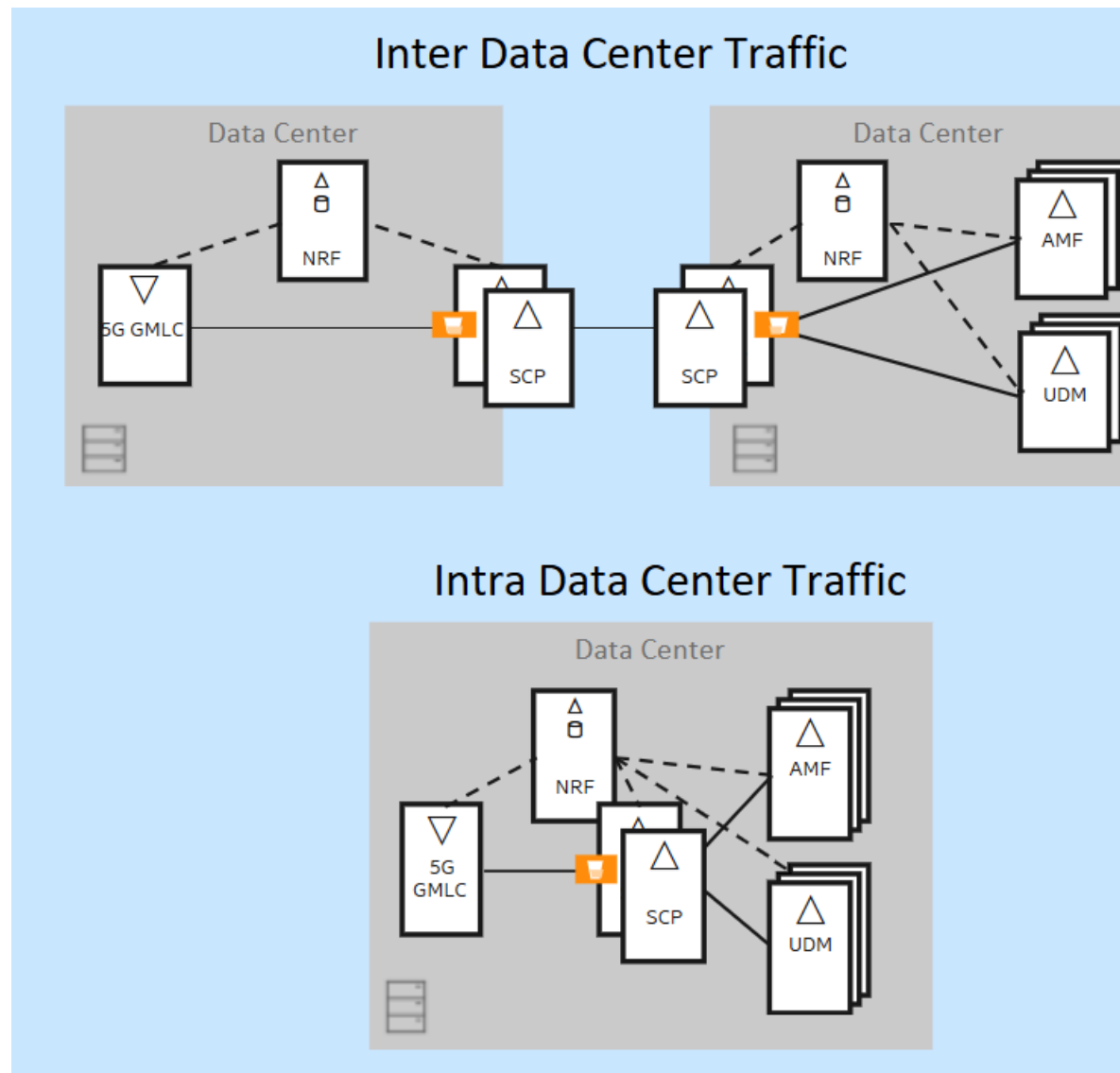


Figure 3 SCP Deployment between NF Consumer and Producer

2.1.1

Feature Overview

The NLG implements the 5G GMLC entity for the 5G LCS architecture and the procedures defined by 3GPP. The 5G GMLC entity is located on the NLG server, and uses one internal supporting function, which is the database for storing NR cells.

The 5G GMLC is capable of handling the following positioning requests:

- current
- last



- last or current
- current or last

For more information, see the *Cached Location* section in Technical Product Description, NLG.

2.2 Signaling Procedures

2.2.1 5GS-MT-LR Standard

The 5G GMLC can serve Mobile Terminating Location Requests (MT-LRs) with a UDM query or without a UDM query (cached AMF ID) over secure or non-secure HTTP2 connection.

For more information, see the *5GS Domain Signaling Procedure* section in Technical Product Description, NLG.

2.2.2 5G Core Procedures

2.2.2.1 NRF

The 5G GMLC supports the following services toward the NRF according to the *3GPP 29.510* TS:

- Nnrf_NFDiscovery
- Nnrf_NFManagement



2.2.2.1.1 NF Discovery

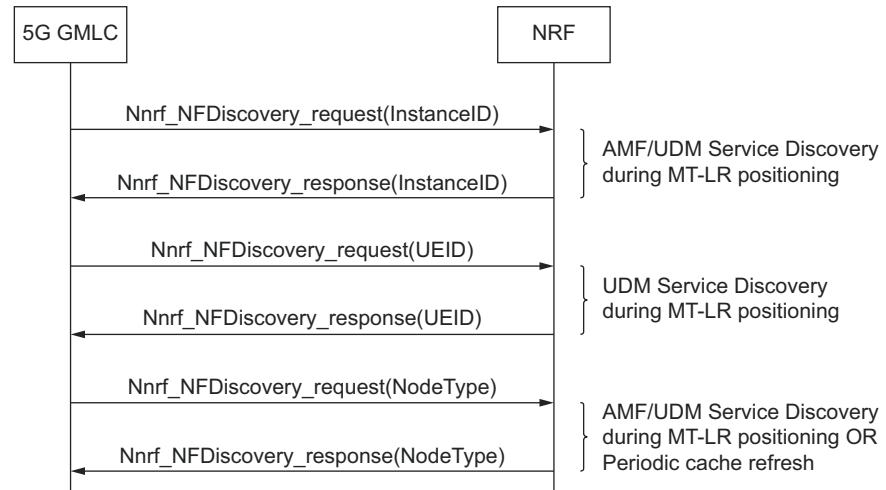


Figure 4 NF Discovery Procedures

The Service Discovery function is used to retrieve information about the UDM or the AMF. In every case, the 5G GMLC sends a GET request over HTTP2 towards one of the configured NRF servers. The syntax of the request is the following:

```
<Server Address><API Prefix>nrf-disc/v1/nf-instances?<Parameter List>
```

The following types of discovery procedures exist:

- Nnrf_NFDiscovery_request(InstanceID)
- Nnrf_NFDiscovery_request(UEID)
- Nnrf_NFDiscovery_request(NodeType)

All the listed discovery procedure paths contain the RequesterNFType parameter, which is always GMLC, and the TargetNFType parameter, and they vary depending on the type of the discovery procedure.

When OAuth 2.0 is configured, a valid access token must be provided in the request header.

For more information on the RequesterNFType and the TargetNFType parameters, see the [Parameter List](#) table in [AMF Provide Positioning Info Query](#) on page 13.

The response to a discovery request is called SearchResult, and it is received in JSON format. The SearchResult contains a validityPeriod integer and an array of NFProfiles called NfInstances.



The validity period is applicable for all the NF instances in the `SearchResult`. The `NfProfiles` contain all required information to send a message to a specific 5G node, such as the IP address and the node type.

Table 1 NfProfiles Object

Object Name	Properties
NfProfile	— InstanceId — NfType — NfServices (array of NfService) — UdmInfo
NfService	— scheme (HTTP and HTTPS) — ipEndPoints (array) — apiPrefix
ipEndpoint	— ipv6Address — ipv4Address — port — TransportProtocol (always TCP)

If the `SearchResult` does not contain any usable information, which means it does not meet the requirements listed in *TS 29.510*, the discovery procedure fails.

Nnrf_NFDiscovery_request (InstanceID)

This procedure is used only during a positioning when the cache does not contain information about the node that is associated with the user equipment, the position of which is requested.

The `TargetNFType` parameter must be set to AMF or UDM.

If `TargetNFType` is set to AMF, the `service-names` parameter must be set to `namf-loc`.

If `TargetNFType` is set to UDM, the `service-names` parameter must be set to `nudm-eucm`.

Since the `TargetNfInstanceId` is a unique identifier, this procedure always returns information about a single serving node. The IP addresses and the API prefix used in the 5G GMLC are the results of this procedure.

Example 1 Nnrf_NFDiscovery_request

```
127.0.0.1:12345/nnrf-disc/v1/nf-instances?target-nf-type=UDM&requester-nf-type=GMLC&service-names=nudm-eucm&target-nf-instance-id=1798ecec-7306-4ab2-9adc-fcb80ae948eb →
```



Nnrf_NFDiscovery_request (UEID)

This procedure is similar to `Nnrf_NFDiscovery_request (InstanceID)`. The difference is that it uses one of the unique IDs given by the LCS client instead of the InstanceID.

The `Target NF Type` parameter is set to UDM. The `service-names` parameter must be set to `nudm-eucm`. The User Equipment Identifier (UEID) parameter is either `supi` or `gpsi`, and corresponding to that, either the `supi` or the `gpsi` parameter is set.

The IP addresses and the API prefix obtained as the result of this procedure are used in the 5G GMLC.

Example 2 Nnrf_NFDiscovery_request (UEID)

```
127.0.0.1:12345/nnrf-disc/v1/nf-instances?target-nf-type=UDM&requester-nf-type=GMLC&service-names=nudm-eucm&supi=imsi-132456978156314 →
```

Nnrf_NFDiscovery_request (NodeType)

This discovery procedure is used for the periodical cache refresh.

The `Target NF Type` parameter must be set to AMF or UDM.

The IP addresses and the API prefix obtained as the result of this procedure are used in the 5G GMLC. If the target NF type is set to UDM, the `UdmInfo` field is also parsed after a successful discovery, and used to build the UDM cache.

The `UdmInfo` contains `supi` ranges and `gpsi` ranges, which determine which user equipments are served by that instance of the UDM.

For more information on the `Target NF Type`, `supi`, and `gpsi` parameters, see the [Parameter List](#) table in [AMF Provide Positioning Info Query](#) on page 13.

For more information on the cache feature, see [Destination Selection](#) on page 32.

2.2.2.1.2 NF Management

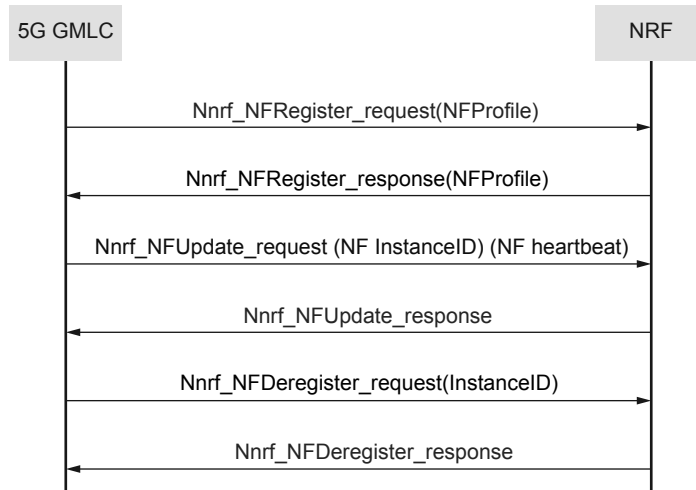


Figure 5 NF Management Procedures

The **Nnrf_NFManagement** service is used for registering the 5G GMLC NF Profile to the NRF server to inform other participants of the network about the notification callback path of the 5G GMLC.

The supported **Nnrf_NFManagement** services are the following:

- **Nnrf_NFRegister**
- **Nnrf_NFUpdate(NF heartbeat)**
- **Nnrf_NFDeregister**

Nnrf_NFRegister

This service is used for registering the 5G GMLC NF profile to an NRF server.

The 5G GMLC sends its NF profile to the NRF server, which then registers and distributes it for the upcoming requests. In case of an unsuccessful registration process, the 5G GMLC retries the process until any of the following occurs:

- The attempts reach the maximum limit.
- The 5G GMLC gets a permanent error response from the NRF server.
- The 5G GMLC gets a timeout.

If more NRF servers are configured and the registration process to an NRF server fails, the 5G GMLC tries to register its NF profile to another available NRF server.



If the primary server becomes available after unreachability, the 5G GMLC tries to shift back to the primary NRF server from another: the 5G GMLC tries to register to the primary server, and if it is successful, then sends a delete request to the previous NRF.

If the registration process is successful, the NRF sends back a HTTP2 201 Created response to the 5G GMLC.

In case of error, the NRF sends HTTP2 error messages (3xx, 4xx, 5xx).

In case the registration to all NRF servers fails, a new alarm is raised. For more information, see [Alarm Descriptions](#) for NLG.

NF Profile

The NF Profile supports the following attributes:

- defaultNotificationSubscriptions (Array)
 - notificationType (LOCATION_NOTIFICATION)
 - callbackUri (the URL and the path where the 5G GMLC can be reached for notification)
- enableOAuth
- fqdn
- gmlcInfo (Enum)
- heartbeatTimer (Heartbeat periodic interval in sec)
- ipv4Addresses
- ipv6Addresses
- nfInstanceId (UUID of the 5G GMLC Node)
- nfProfileChangesSupportInd (Bool)
- nfStatus (REGISTERED)
- nfType (GMLC)

Parameters

For the communication parameters used between the 5G GMLC and the NRF server, see the *Http2* section in [Parameter List](#) for NLG.

Nnrf_NFUpdate (NF heartbeat)

The Heartbeat messages inform the NRF server that the 5G GMLC is still alive, functioning, and accepting communication. Heartbeat messages are periodically sent by the 5G GMLC toward the NRF. The NRF acknowledges them by sending back a HTTP2 200 OK message with the NFProfile, or a HTTP2 204 No Content message. The NRF can override the default notification period during the registration process and the 5G GMLC follows that request. If there is no response for three consecutive times, or if an error arrives from the NRF as a response, the 5G GMLC classifies the NRF server as unreachable and tries to register to an available NRF.

Nnrf_NFDeregister

If the Location Notification service is no longer available on the 5G GMLC, or if the 5G GMLC is shifted to another NRF, then an NF profile deregistration request is sent to the corresponding NRF.

When OAuth 2.0 is configured, the NFDeregister header must include a valid access token.

2.2.2.2

UDM

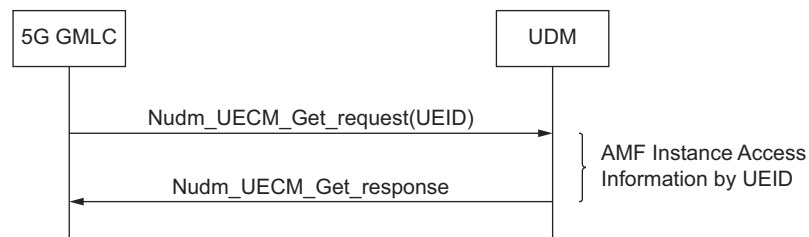


Figure 6 UDM UECM Get Query

In 5G GMLC, the User Equipment Context Management Get (UECMGet) request is used to retrieve information about the AMF instance, which serves the user equipment under positioning.

The GMLC sends out the request to the following path:

```
<ServerAddress><API Prefix>nudm-uecm/v1/VarUeId/registrations/
amf-3gpp-access.
```

The response to this request contains an Amf3GppAccessRegistration object in JSON format.

The Amf3GppAccessRegistration contains various information about a specific AMF, but the 5G GMLC only uses the following fields of it:

- amfInstanceId (mandatory): If this instance ID is present in the cache, the GMLC uses this to forward the positioning towards the AMF. If this instance



ID is present in the cache, it is used. If the ID is not present, a discovery procedure is triggered.

- `pei` (optional): The request towards AMF can use the `pei` as a path parameter.
- `supi` (mandatory)
- `purgeFlag` (optional): This flag indicates whether the AMF specified with the `AmfInstanceId` stopped serving the UE specified by the UEID.

If any of the mandatory fields is missing, the flow is canceled, and the 5G GMLC responds with an error message.

For more information on the `ServerAddress`, `amfInstanceId`, `pei`, and `supi` parameters, see the [Parameter List](#) table in [AMF Provide Positioning Info Query](#) on page 13.

When OAuth 2.0 is configured, a valid access token must be provided in the request header.

2.2.2.3 AMF

When OAuth 2.0 is configured, a valid access token must be provided in the request header.

2.2.2.3.1 AMF Provide Positioning Info Query

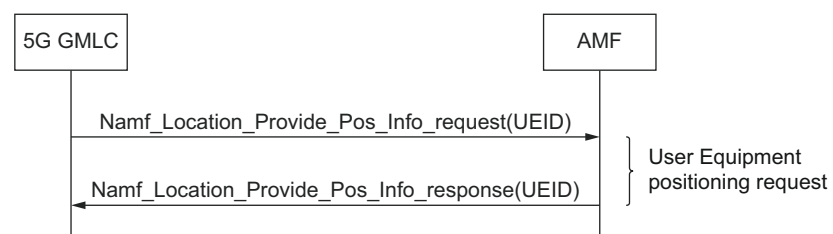


Figure 7 AMF Provide Positioning Info Query

In 5G GMLC, the Provide Positioning Info (PPI) is used to retrieve information about a specific user equipment location. The GMLC sends out the request to the following path:

```
<ServerAddress><API Prefix>namf-loc/v1/<UeContextId>/provide-pos-info
```

For more information on the parameters, see [Table 2](#).

Attention!

If the LCS Client sent the MSISDN of the user equipment, and the response to the UDM request did not contain the Permanent Equipment Identifier (PEI) of the user equipment, the PPI request cannot be sent, and the NLG sends back an error message to the LCS client, because the UeContextId is either a Subscription Permanent Identifier (SUPI) or a PEI. A SUPI can be made if the LCS sends an IMSI. From UDM 3GPP 29.503 v16.8, this problem is resolved, since the Nudm_UECM_Get(Amf3GppAccessRegistration) response can contain SUPI. For more information, see [Statement of Compliance to 3GPP TS 29.503 - 5G System; UDM Services](#).

With the PPI request, the 5G GMLC sends a JSON payload, RequestPosInfo. This payload can contain the following information:

- lcsClientType (mandatory): The type of the LCS client that initiated the positioning request, for example emergency, lawful intercept or commercial.
- lcsLocation (mandatory): The type of the positioning request, for example, current or last.
- Priority (mandatory): The priority of the positioning request.

For more information about the parameters above and additional information about the request, see [3GPP TS 29.518](#).

The ProvidePosInfo response to the PPI request arrives to the 5G GMLC in a JSON format, which contains the location estimate of a specific user equipment and other important information. The location estimate itself contains a shape, which contains coordinates and additional information of the shape, depending on the shape type.

There are two more important elements that can be present in the response JSON:

- ncgi (optional): Information about the NR Cell where the user equipment is located. This can be translated into a location estimate.
- positioningDataList (optional): The list of positioning techniques used during the positioning of the user equipment.

If ncgi and location estimate tag are missing, the 5G GMLC responds with an error message. For more information about the response, see [3GPP TS 29.518](#).

Table 2 NF Service Properties

Name	Format	Description
Server Address	^IPv4 IPv6:PORT\$	The address of the 5G Core node.



Name	Format	Description
API Prefix	^/.*\$	Optional deployment specific string, which has to start with a „/“ character.
IPv4	'^((([0-9] [1-9][0-9] 1[0-9][0-9] 2[0-4][0-9] 25[0-5])\.){3}([0-9] [1-9][0-9] 1[0-9][0-9] 2[0-4][0-9] 25[0-5]))\$'	An address that follows the IPv4 protocol. The format is defined in <i>TS 29.571</i> .
IPv6	- ^((: 0? ([1-9a-f][0-9a-f]{0,3}))*)(0? ([1-9a-f][0-9a-f]{0,3}))*{0,6}(: 0? ([1-9a-f][0-9a-f]{0,3}))*\$ - ^(((^[^:]+):){7}([^[^:]+)) (((^[^:]+)*[^[^:]+)?::((^[^:]+):){1-6})\$	An address that follows the IPv6 protocol. The format is defined in <i>TS 29.571</i> .
PORT	min: 0 max: 65535	The port of the service, which is defined in <i>TS 29.510</i> .

Table 3 Parameter List

Name	Format	Description
TargetNfType	target-nf-type=<NFType>	The type of the NF that the request is sent to.
NFType	^(NRF UDM AMF)\$	One of the node types that the 5G GMLC communicates with. This is a narrowed down list from the standard. <i>TS 29.510</i> contains the full list.
RequesterNfType	requester-nf-type=GMLC	The NF type of the request sender. Default value is GMLC and it is non configurable.
TargetNfInstanceId	target-nf-instance-id=<UUID>	The ID of the NF instance as defined in <i>TS 29.571</i> .
UUID	^[0-9A-F]{8}-[0-9A-F]{4}-[4][0-9A-F]{3}-[89AB][0-9A-F]{3}-[0-9A-F]{12}\$	Universally Unique Identifier. The version is 4.
UeContextId	^(supi pei)\$	User equipment context identifier
VarUeId	^(supi gpsi)\$	The amf3gppregistration request path parameter.
supiParameter	supi=<supi>	The supi parameter of a request path.
gpsiParameter	gpsi=<gpsi>	The gpsi parameter of a request path.
peiParameter	pei=<pei>	The pei parameter of a request path.
supi	^(imsi-[0-9]{5,15} nai-.+ .+) \$	The SUPI of a specific user equipment.
gpsi	^(msisdn-[0-9]{5,15} extid-[^@]+@[^@]+ .+) \$	The Generic Public Subscription Identifier (GPSI) of a specific user equipment.
pei	^(imei-[0-9]{15} imeisv-[0-9]{16} .+)\$	The PEI of a specific user equipment.
amfInstanceId	UUID	Instance ID of a specific AMF node.
ratType	^(NR EUTRA WLAN VIRTUAL)\$	Radio Access Technology Type

2.2.2.3.2 AMF Event Notify

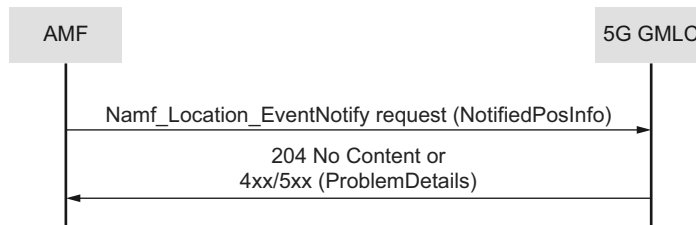


Figure 8 AMF Event Notify

The 5G GMLC accepts notifications from the AMF on an `Namf_Location_EventNotify` API about positioning results performed at the emergency call setup of the UE.

The default API URI is as follows:

```
http(s)://<NLG Server Address>/gmlc/namf-loc/v1/EventNotify
```

On the `Namf_Location_EventNotify` API, the AMF sends a `NotifiedPosInfo` message to the GMLC with a JSON payload containing the information processed by the GMLC:

- `locationEvent`
- `supi`
- `gpsi`
- `pei`
- `locationEstimate`
- `positioningDataList`
- `gnssPositioningDataList`
- `ncgi`
- `servingNode`

For more information on the parameters and additional information about the request, see *3GPP TS 29.518*.

The response to the `NotifiedPosInfo` message can be either one of the following:

- **204 No Content response:** If the received `NotifiedPosInfo` is correct and the GMLC server is able to process it.



- Negative 4xx/5xx response: If either the request is incorrect or an error occurs during the processing of the message.

In case of a negative response, an optional additional `ProblemDetails` JSON payload can indicate the reason of the error.

In case no error occurs during the processing of the `NotifiedPosInfo` message, the GMLC can send the location information to the configured LCS client. Sending the EMEREP can be configured by the `EnableFirstPush` parameter. The content of the request is also saved into the internal NI-LR cache and the subscriber database.

2.2.3 5G GMLC Indirect Communication

Operations

ENL supports indirect communication Model C options 1 and 2, as follows:

Indirect communication Model C option 1:

In active indirect communication scenarios, the nodes of the 5G GMLC is using the `3gpp-sbi-target-apiRoot` header in the request. In case of erroneous responses for existing operations, the `Server` or `Via` headers are added. The operations are as follows:

- `Amf3GppAccessRegistration` (NLG > UDM): through the **Nudm_UEContextManagement** service
- `ProvidePositioningInfo` (NLG > AMF): through the **Namf_Location** service
- `EventNotify` (AMF > NLG): through the **Namf_Location** service

Indirect communication Model C option 2:

If Model C option 2 is enabled, apart from the scenarios of Model C option 1, the GMLC includes the following headers in the requests for target NF producer reselection by the SCP:

- `3gpp-Sbi-Discovery-target-nf-set-id` contains the target NF Set ID corresponding to the target NF selected by the NLG.
- `3gpp-Sbi-Discovery-target-nf-service-set-id` contains the target NF Service Set ID corresponding to the target NF selected by the NLG.

Apart from the above, GMLC can parse the `3gpp-Sbi-Producer-Id` header if present in the service response (200, 4XX, 5XX) from the SCP. This information is used to populate position record information for GMLC position requests.

Proxy Provisioning

The addresses of the SCP nodes available in the network are provisioned in the NLG. It is possible to configure more SCP instances.

An SCP is configured using its FQDN, consisting of the scheme, authority and API prefix parts.

For more details, see the Related Information at the bottom of the section.

Behavior Model

A dedicated administration state switch was introduced to enable or disable the indirect communication approach. For more details, see the Related Information.

The NLG performs round-robin load sharing among the SCP instances, and maintains dual TCP sessions toward every provisioned SCP instance.

The NLG supports SCP monitoring functionality using the HTTP/2 PING frame with configurable interval between consecutive pings.

- Note:**
- In "mixed communication cases", the communication between the NLG and UDM and between the NLG and AMF is asymmetric, with the parties separately involving or leaving the SCP out from the message handling.
 - The communication between the NLG and the NRF remains direct.
 - It is not possible to selectively activate indirect communication for different 5G services. For example, it is not possible for **Nudm_UEContextManagement** to have indirect communication activated while it is not activated for **Namf_Location**.
 - If Model C option 2 is enabled and NRF does not send `nfSetIdList` and `nfServiceSetIdList` during NF Discovery, then GMLC will fall back to Model C option 1.
 - There is no fallback to direct communication if all SCPs fail.

— RELATED INFORMATION —

Parameter List

OAM REST API v1.0 Specification

2.2.3.1

Generic 3GPP Indirect Communication Flow

The generic flow of the 3GPP indirect communication is as follows:



1. NF target selection by the service consumer, based on cache or NF discovery.
2. SCP selection in a round-robin manner by the service consumer from the provisioned available SCPs.
3. Service request sending by the service consumer to the SCP. The target URI is composed of the SCP API root and the service producer API path.
 - The service consumer also adds the following additional headers to the request:
 - `3gpp-sbi-target-apiRoot` contains the `apiRoot` of the service provider.
 - Authorization is included if OAuth2 is enabled.
 - If the 5G Indirect Communication Model C option 2 is enabled, the service consumer also adds the following headers to the request along with the headers above:
 - `3gpp-Sbi-Discovery-target-nf-set-id` contains the target NF Set ID.
 - `3gpp-Sbi-Discovery-target-nf-service-set-id` contains the target NF Service Set ID.
 - NF consumer adds this information from the cached `nfSetId` and `nfServiceSetId` received during NF producer discovery, according to the conditions below:
 - If the NRF does not send `nfSetIdList` and `nfServiceSetIdList` during NF Discovery, then the service request towards SCP will not include the `3gpp-Sbi-Discovery-*` headers.
 - If any or both of `nfSetIdList` and `nfServiceSetIdList` are present along with `plmnList` configuration (NF Profile configuration parameter of HTTP2 component) and if no `setIds` match the PLMN ID configured in `plmnList`, a fallback will happen to Model C option 1, and the service request will not include `3gpp-Sbi-Discovery-*` headers.
 - If `plmnList` is empty, and if any or both of `nfSetIdList` and `nfServiceSetIdList` are present in the Discovery response, then the first entry (`setId`) of the list is selected.
 - If there are multiple `setIds` matching with the same PLMN, then the first matching `set id` is selected from the `nfSetIdList` and `nfServiceSetIdList`.
 - The `3gpp-Sbi-Discovery-*` headers can be used by the SCP to reselect the target NF producer, in case the NF producer instance selected by the NF consumer cannot serve the request.

4. Request forwarding to the service provider by the SCP.
 - Replacing the SCP `apiRoot` with the contents of the `3gpp-sbi-target-apiRoot`.
 - Removing the `3gpp-sbi-target-apiRoot` header.
5. Response construction in the NF Producer.
 - Addition of a `Server` header in case of erroneous response.
6. Response forwarding to the service consumer by the SCP.
 - In case of NF producer reselection, the SCP sends the service response to the NF consumer with the `3gpp-Sbi-Producer-Id` header, which contains any or all of `instanceId`, `serviceInstanceId`, `nfSetId`, and `nfServiceSetID` of the NF producer.
 - If an error occurs in the SCP, it adds a `Server` header in the created erroneous response.
 - If an error occurs and the service provider included a `Server` header in the response to the SCP, the SCP copies that response to the service consumer.
 - If the SCP gets an error response from the service producer that does not include a `Server` header, a `Via` header is set.
7. Determining the error source by the service consumer.
 - Trying with the new SCP if the source was the immediate SCP.

2.2.3.2 Basic Indirect Operation Flow Handling

Positive MT-LR Flow

Figure 9 indicates the difference between the direct and indirect communication 5G MT-LR flows.

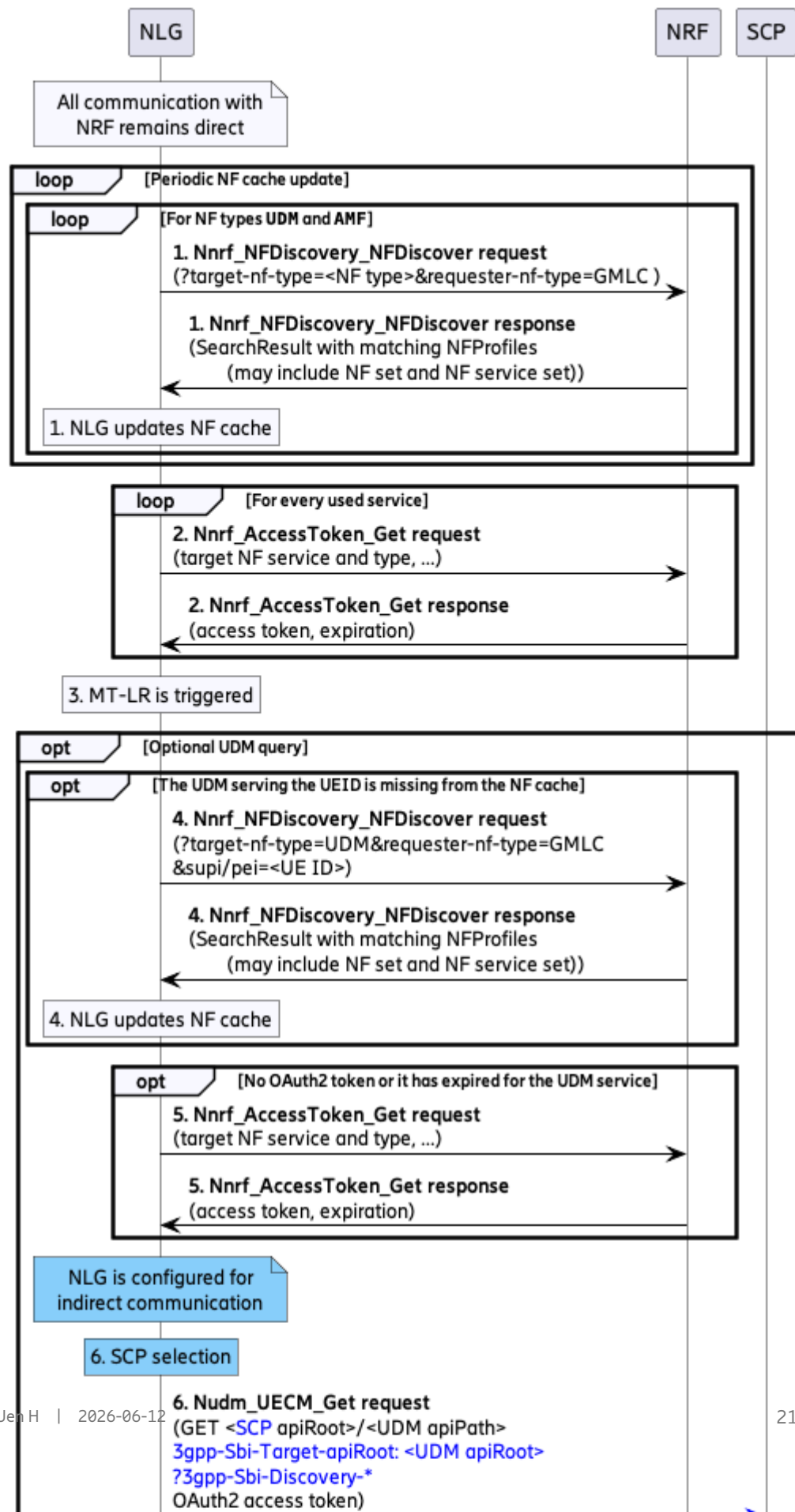


Figure 9 Positive MT-LR Flow

- Steps 1 through 5 are identical with the direct communication flow, but the NF set ID and NF service set ID are used only in the indirect communication.
- In Step 6, the NLG sends the Nudm_UECM_Get request, but the target address of the request is one of the addresses of the SCP node, selected from the provisioned list.

The 3gpp-Sbi-Target-apiRoot: *<UDM ApiRoot>* additional HTTP header then guides the SCP to replace the specified SCP apiRoot with the UDM apiRoot. The SCP then forwards the modified request to the UDM.

If indirect communication option 2 is enabled, the NLG also sends 3gpp-Sbi-Discovery-* headers, and receives the 3gpp-Sbi-Producer-Id header corresponding to the NF provider reselection done by the SCP.

- Steps 7 and 8 remain identical with the direct communication flow.
- In Step 9, the NLG sends the Namf_Location_ProvidePositioningInfo request to the same SCP instance as in Step 6, with the 3gpp-Sbi-Target-apiRoot: *<AMF ApiRoot>* additional HTTP header.

The header value makes the SCP replace the original *<SCP apiRoot/AMF apiPath>* parameter of the GET request with *<AMF apiRoot/AMF apiPath>*.

If indirect communication option 2 is enabled, the NLG also sends 3gpp-Sbi-Discovery-* headers, and receives the 3gpp-Sbi-Producer-Id header corresponding to the NF provider reselection done by the SCP.

- At the end of the process, the SCP node forwards the response to the NLG.

Positive NI-LR Flow

Figure 10 depicts the flow with an incoming emergency event from the AMF through the SCP.

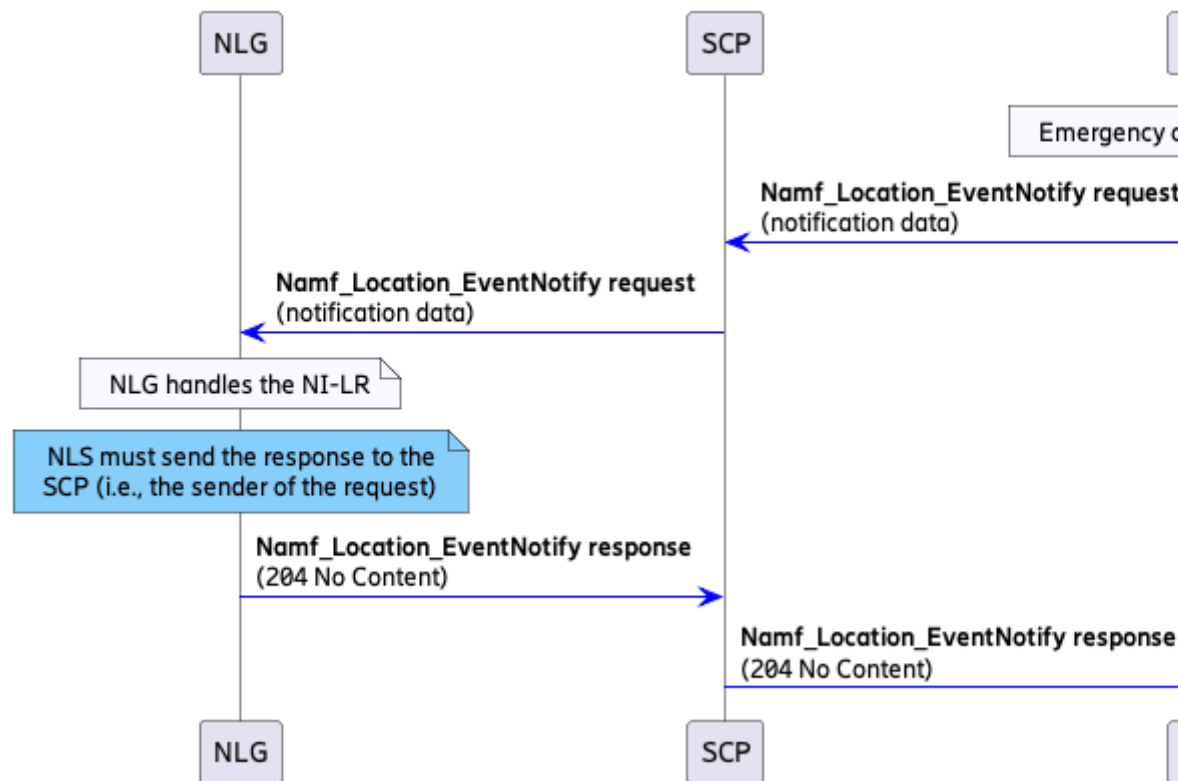


Figure 10 Emergency NI-LR Flow

In this case, the NLG sends the `Namf_Location_EventNotify` response with a 204 response code to the same SCP instance rather than randomly choosing one from the provisioned list.

Mixed Communication: NLG Configured for Direct Communication

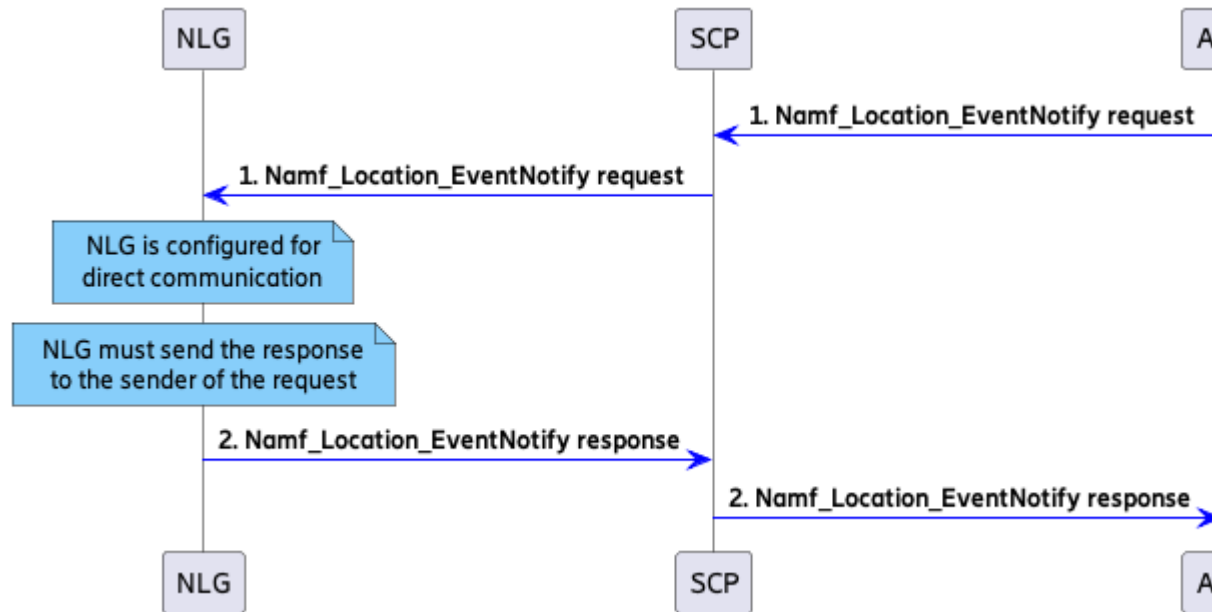


Figure 11 NLG Configured for Direct Communication

In this case, the NLG sends the `Namf_Location_EventNotify` response with a 204 response code to the same SCP instance, even if the communication is configured to be direct.

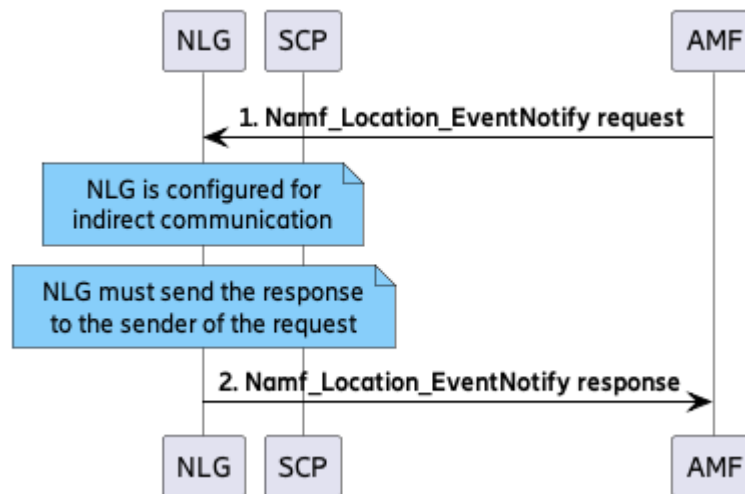


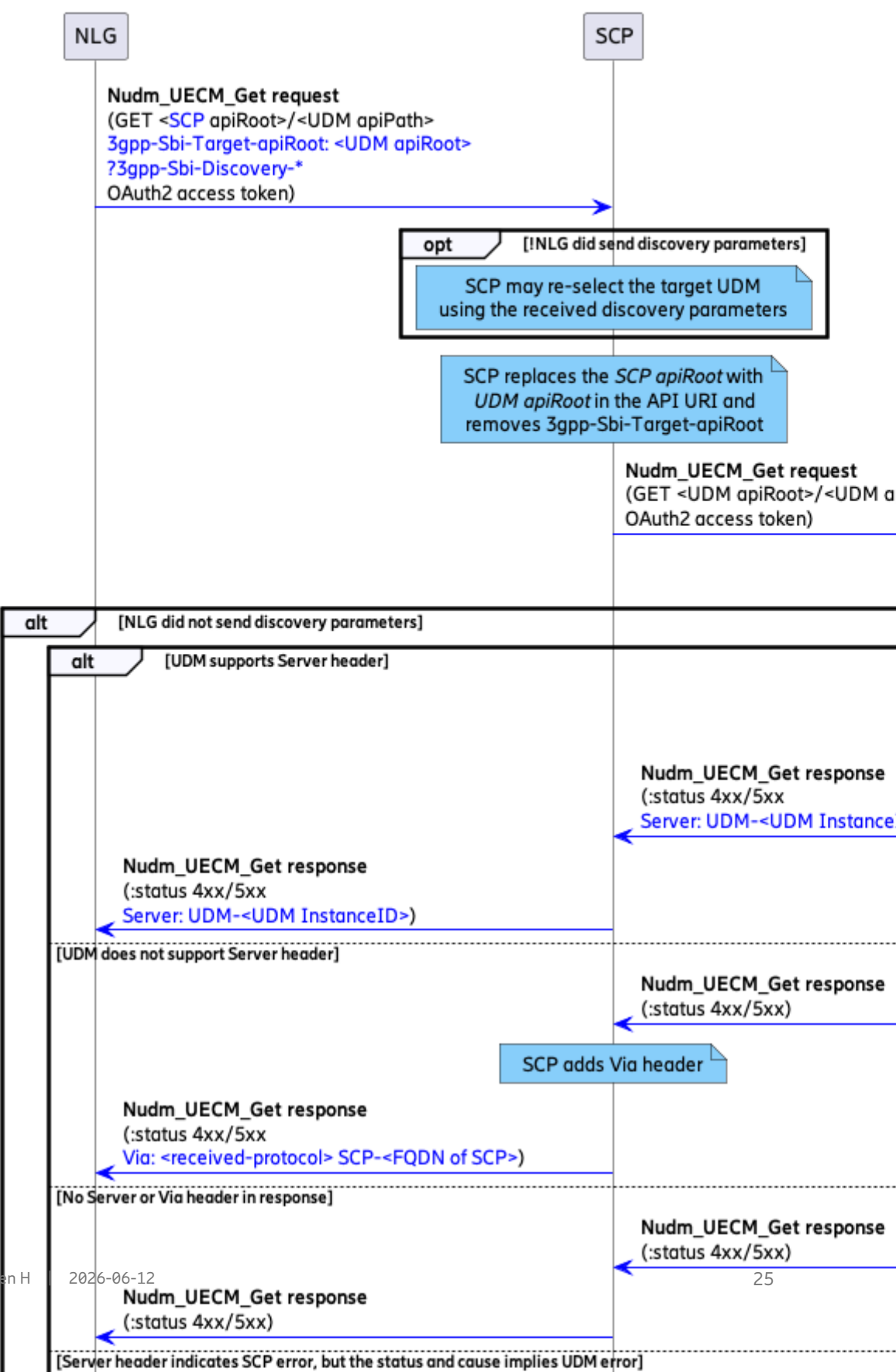
Figure 12 NLG Configured with Indirect Communication

Despite being configured with indirect communication, the NLG replies to direct requests with direct answers without turning to any provisioned SCP instance.

2.2.3.3

Error Handling during Indirect Communication

Producer Errors on the UDM and AMF Side



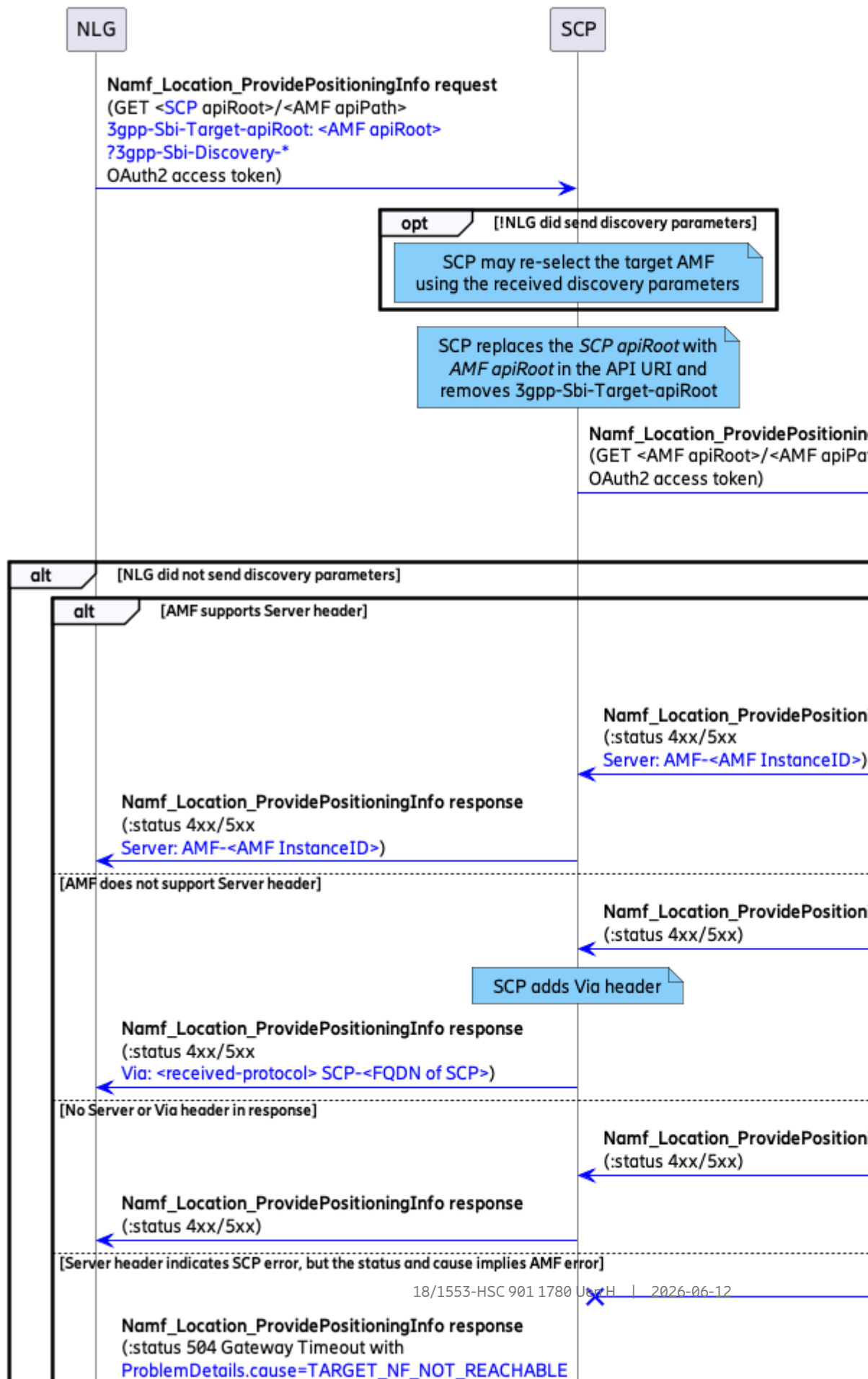


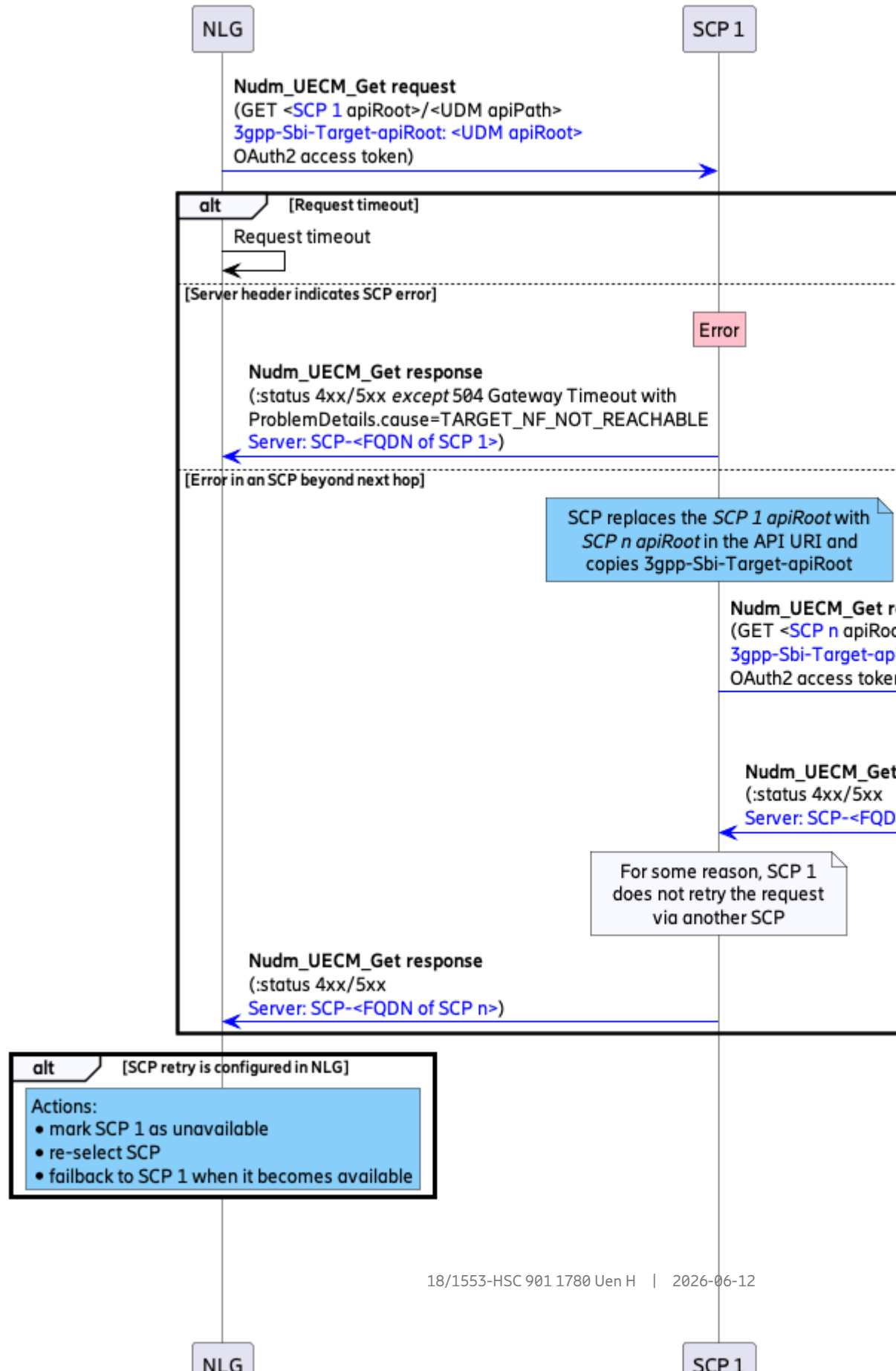


Figure 14 AMF Side Errors

In this case, the UDM and AMF act as producers, and the NLG acts as a consumer. The NLG can be notified about errors in the following ways:

- The producers add a `Server` header in the 4xx/5xx responses.
- The producers skip adding the `Server` header. In this case, the SCP adds a `Via` header to the response, including its own FQDN.
- Neither the producers, nor the SCP populate these headers.
- The SCP returns a 504 Gateway Timeout response, specifying `TARGET_NF_NOT_REACHABLE` as the cause and the SCP identifier in the `Server` header.
- In case of option 2, if an NF producer error happens and the NLG did send discovery headers in the request to the SCP, the SCP may resend the request to another NF producer according to the discovery parameters provided by the NLG.

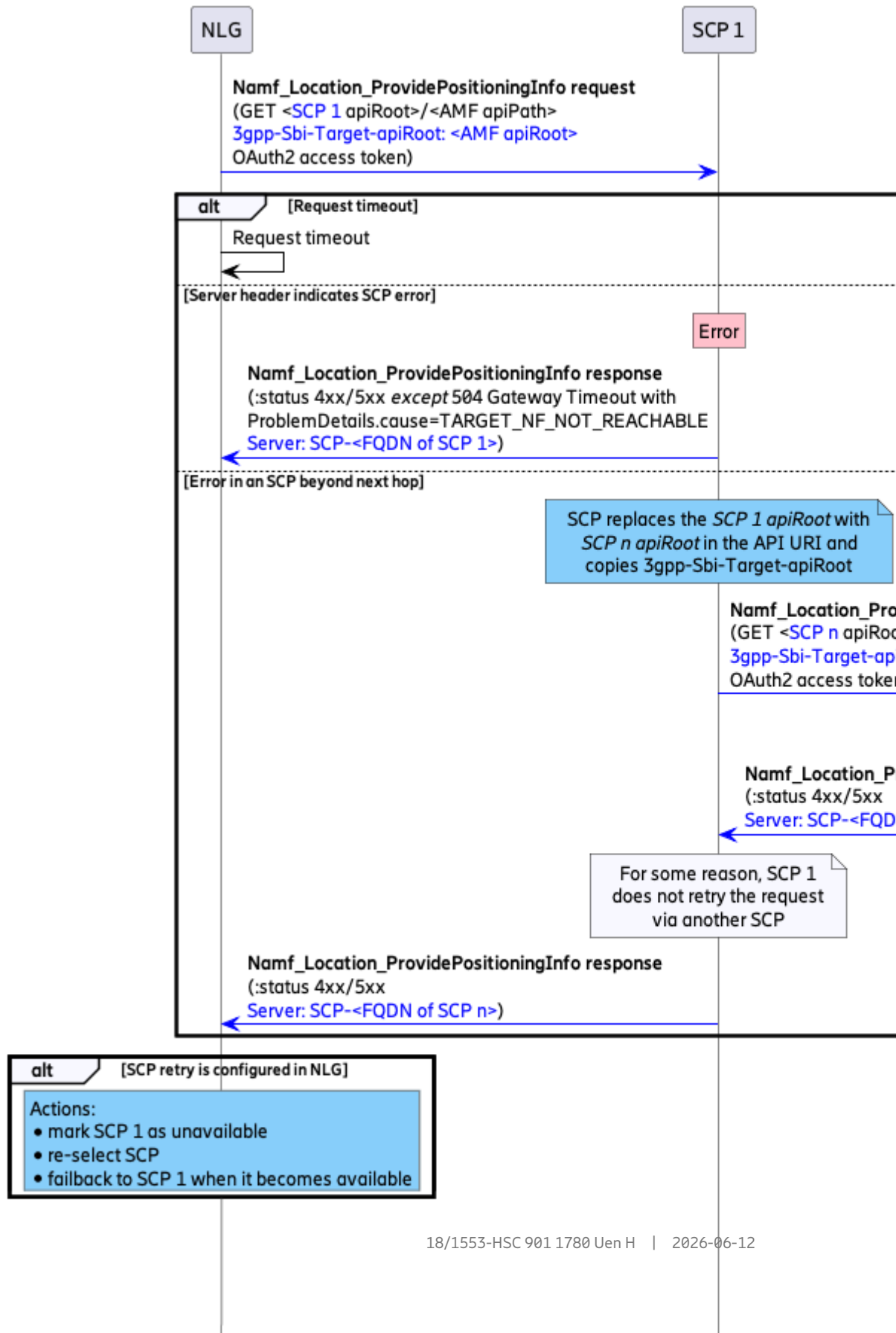
Example 3 SCP Error During Communication with UDM





SCP Error During Communication

Example 4 SCP Error During Communication with AMF





The NLG treats the following cases as an SCP error:

- No response from the serving SCP during indirect communication between the NLG and AMF.
- A `Namf_Location_ProvidePositioningInfo` response with a 4xx/5xx (with the exception of 504 together with `TARGET_NF_NOT_REACHABLE` specified as a cause) status and the problematic SCP identifier.
- The chosen SCP does not retry forwarding the request to any other SCP and sends back a 4xx/5xx `Namf_Location_ProvidePositioningInfo` response with a `Server` header containing the problematic SCP identifier.
- No response from the serving SCP during indirect communication between the NLG and UDM.
- A `Nudm_UECM_Get` response with a 4xx/5xx (with the exception of 504 together with `TARGET_NF_NOT_REACHABLE` specified as a cause) status and the problematic SCP identifier.
- The chosen SCP does not retry forwarding the request to any other SCP and sends back a 4xx/5xx `Nudm_UECM_Get` response with a `Server` header containing the problematic SCP identifier.

NLG Internal Issues during Communication

If the NLG cannot provide a proper response to the incoming NI-LR requests due to internal issues, it will continue to follow the same communication directives as in the proper cases:

- In case of internal issues upon an incoming `Namf_Location_EventNotify` request from an SCP instance, the response will be sent back to the same SCP instance.
- The NLG sends the `Namf_Location_EventNotify` response with a 4xx/5xx status code and the sender NLG identifier in the `Server` header.
- The SCP forwards the response to the AMF that sent the request.



2.3 5G GMLC Features

2.3.1 Parameter Settings

Connection Establish Timeout

This parameter controls for how long the 5G GMLC waits for an AMF or UDM connection to be established. The default value of this parameter is 100 ms, which can be adjusted if the NLG is installed in a high latency network.

The parameter can be configured only through the **OAM Rest** interface. For more information about the **OAM Rest** interface, see [OAM REST API v1.0 Specification](#).

For the description of the ConnectionEstablishTimeout parameter, see the *Http2* section in [Parameter List, NLG](#).

Secondary NRF Parameters

The 5G GMLC can function with one NRF configured, secondary NRF parameters are not required.

To establish a connection towards only one NRF in the network, the primary NRF IPv4, or IPv6 parameter, and the Port parameter must be filled properly, and the secondary NRF IPv4, IPv6, and Port parameters left empty.

For the NRF parameters, see the *Http2* section in [Parameter List, NLG](#).

2.3.2 Destination Selection

During the positioning of a UE, the 5G GMLC needs to determine to which UDM and AMF it sends the request. There are the following major ways to gather this information:

- The periodically refreshed cache contains the address of the needed 5G Core node.
- Discovery procedures. For more information, see [NRF Destination Selection](#) on page 33.

UDM

1. The GMLC checks if it has InstanceID or UEID information.
2. If the GMLC has InstanceID or UEID information, it queries the cache with the given information. The InstanceID is preferred.

Note: The InstanceID can be configured in the Network Data section on the GUI and the UEID comes from the LCS client.



3. If the query is successful, the GMLC queries the retrieved node.
4. If the query is unsuccessful, the GMLC starts a discovery procedure.
Note: The UDM can be discovered both by InstanceID and UEID.
5. If the discovery procedure fails with either InstanceID or UEID, the GMLC sends back an error message.

The NLG selects the first matching UDM from the UDM cache. If the UDM cache is updated, another UDM might be selected, as the order of the UDMs in the cache is undetermined. This means that the NLG does not apply load balancing among matching UDMs, but the same UDM is selected for the positioning procedures. The selected UDM can change at cache update.

AMF

1. The GMLC checks if it has InstanceID.
2. If the GMLC has InstanceID, it queries the cache with the InstanceID.
3. If the query is successful, the GMLC queries the retrieved node.
4. If the query is unsuccessful, the GMLC starts a discovery procedure.
5. If the discovery procedure fails, the GMLC sends back an error message.

Note: The 5G GMLC always queries the first NF instance returned by the NRF and stored in the cache that matches the query criteria.

2.3.3

NRF Destination Selection

The 5G GMLC determines the availability of the configured NRF servers by checking if it can connect to them or not. If an NRF server address is configured properly on the GUI, the 5G GMLC attempts to connect to it until the configuration is changed, or until the 5G function or the HTTP2 component is shut down. If one or both of the NRFs are unavailable, an event and an alarm are raised containing information about the unavailable NRFs.

The two configured NRF servers work as an active-standby, and only one is used actively for queries. By default, the primary NRF server is used actively for query.

The configured NRFs can have both an IPv4 and an IPv6 address configured, see [IPv4](#), [IPv6](#) on page 34.



2.3.4 IPv4, IPv6

The 5G GMLC can communicate on both IPv4 and IPv6 with the 5G core nodes. 5G GMLC supports fallback through preferred protocol version if the node is unavailable, and the secondary protocol address is present.

Fallback to the not preferred IP version is only supported during connection creation time. If any of the AMF, UDM, or NRF requests failed, IP version fallback is not supported.

If the connection cannot be established with an IP address with a preferred IP version type, then the 5G GMLC attempts to establish the connection two more times. After the third unsuccessful attempt, if an IP address with a not preferred version type is present, the 5G GMLC tries to connect to that IP address three times. If all three attempts fail, the 5G GMLC falls back to the original IP address, and tries to connect to that IP address.

In case of NRF, the 5G GMLC tries to reconnect until the configuration is changed (modified or the given IP address is removed), or the 5G functionality of the component is turned off.

In case of AMF or UDM, the 5G GMLC tries to reconnect until the given connection is deleted from the cache, or the 5G functionality of the component is turned off.

The preference can be set by the *IP Version Preference* parameter. For more information, see the *Parameters for System Settings in Cluster Level (General Configuration)* table in the *System Settings* section in *Parameter List*, NLG.

2.3.5 NCGI Translation

In case of SLIS/EMELIR positioning, if the AMF returns an NCGI, it is included in the response message towards the client.

If the AMF returns an NCGI only and does not return a location estimate, the 5G GMLC can translate the NCGI into a location estimate; see the *5GS-MT-LR Standard* section in *Technical Product Description*, NLG.

NCGI translation is disabled by default. To enable the feature, see the *PProuter* section in *Parameter List*, NLG.

2.3.6 Ping

When the 5G GMLC successfully establishes a connection towards an NF, the 5G GMLC starts pinging the NF using the HTTP2 ping frame.

The 5G GMLC sends the ping message to the NF every 60 seconds to keep the connection alive when no request is sent on that connection, and to detect if the NF closed the connection gracefully or not.



If the ping detects that the connection is closed, or the ping times out towards either NRFs, the 5G GMLC tries to reestablish the connection.

If the ping fails towards any NFs stored in the cache, the NF is discarded from the cache.

2.3.7

Cache

The 5G GMLC implements a caching function for the connections towards the serving UDM and AMF network functions that are discovered periodically by querying the NRF over the Nnrf SBI.

The period of the cache refresh is set by the validity period returned by the NRF.

Note: If all requests can be served by the NFs already stored in the cache, the 5G GMLC does not send any queries or requests towards the NRF until the validity period timer expires. If the NRF is updated by either the AMF or the UDM during the cache validity period, it is possible that the cache content becomes obsolete, as the 5G GMLC acts as a client only towards the NRF, and does not get notified of the changes.

Request Timeout

When 5G access is enabled (through the GUI under **Networking Connections > Mobile Core Network > AMF/UDM > Access HTTP2**), an initial discovery is requested immediately when the connection with at least one NRF is established. If the NRF responds with the proper information, the next query time is determined by the end of the validity period sent back in the response.

If the NRF fails to respond within the discovery time, the request is disposed. (The discovery time is configured through the GUI under **Software Management > NLG Components > Configuration > HTTP2 > Discovery time**.) The 5G GMLC tries to send the request two more times with discovery time. If all attempts time out, the 5G GMLC schedules a new request 60 seconds later to prevent the NLG from flooding the NRF with requests.

Note:

- After the first request timeout, the 5G GMLC clears the corresponding Network Function cache to avoid using potentially outdated connections.
- If any of the on-demand discovery requests (NFDDiscovery triggered by individual LCS request) succeeds between the first attempt to query the NRF with NFDDiscovery service and the request scheduled 60 seconds later, a new Network Function service, that is not yet saved in the 5G GMLC cache, is stored.

Graceful Cache Clear

During the graceful cache cleanup, connections, that are planned to be discarded from the cache, either wait for the ongoing requests to receive their responses, or terminate immediately after a set timeout to ensure connection shutdown. For

connection termination, the corresponding Network Function request timeout is used.

If connection is lost to all configured NRFs, the cache is cleared.

When the validity period expires, the NLG sends the cache refresh request to the NRF. Then, the following scenarios decide the next operation on the cache:

- The NRF responds with at least one network function. The response is compared to the currently cached connections, and the ones that are not included in the response are discarded from the cache.
- The NRF responds with an error, and the corresponding network function cache is cleared gracefully.
- The request towards the NRF times out. For more information about request timeout, see [Request Timeout](#) on page 35.

Note: Currently running requests are not interrupted by deleting the mentioned connections from the cache.

The NFDISCOVERY result is parsed if the NRF responds with one, and compared to the current cache content. The connections included in the response are kept, while the connections that are not returned are discarded.

Immediate Cache Clear

If 5G access is disabled, the cache is cleared immediately, without letting the ongoing requests wait for their responses from the 5G Core Network Functions.

Cache Features

UDM and AMF caches within the NLG can store connections towards the Network Function services in various ways.

UDM Cache Storage Connection Storing Capabilities

- Corresponding UDM instance ID. (This type of UE correlation needs NLG Network Storage configuration.)
- The GPSI of a UE correlated to a numeric range that determines a group of serving UDMs.
- The SUPI of a UE correlated to a numeric range that determines a group of serving UDMs.

Note: GPSI and SUPI ranges are obtained from the NRF during a periodic or an on-demand NFDISCOVERY that is stored in the UDM cache. If no ranges are returned for a UDM instance, that indicates that the specific UDM can serve any SUPI or GPSI.



AMF Cache Storage Connection Storing Capabilities

- Instance ID of the corresponding AMFs

Note: The NLG obtains the serving AMF service instance ID for a UE from the UDM.

Cache Maintenance

If the HTTP2 connection towards a stored NF fails or times out, the NF is discarded from the cache.

Note: If a request fails with an HTTP2 response from the NF (for example, 404, 500), it remains in the cache, as the connection is still valid.

2.3.8 Fallback

In case the 5GS-MT-LR procedure fails due to AMF query timeout or with an AMF error code that equals a configured set of error codes, the NLG tries the CS positioning if possible.

2.4 Interfaces

The SBIs in the 5GC network use HTTP/2 with JSON as the application layer serialization protocol. The 5G GMLC uses the SBIs listed below for control plane positioning.

Note: To maximize the throughput (TPS) on the interfaces, set the `max concurrent streams` parameter to **20 000** on all NRF, UDM, and AMF servers. This configuration controls the number of requests and responses that can be sent in parallel on one connection.

If the `max concurrent streams` parameter is set too low, the 5G GMLC performance can decrease significantly.

AMF services

The 5G GMLC implements the NF service consumer for the `Namf_Location` service per *TS 29.518*.

Service Name	Service Operations	Operation Semantics
Namf_Location	Request	Request/Response

NRF services

The 5G GMLC implements the NF service consumer for the `Nnrf_NFDiscovery` service per *TS 29.510*.



Service Name	Service Operations	Operation Semantics
Nnrf_NFDiscovery	Request	Request/Response
Nnrf_NFManagement	Request	Request/Response

UDM services

The 5G GMLC implements the NF service consumer for the Nudm_UECM service per *TS 29.503*.

Service Name	Service Operations	Operation Semantics
Nudm_UECM	Request	Request/Response

2.5 Supported MLP Error Codes

When an NF node sends a negative response to the GMLC during positioning, which makes the positioning unsuccessful, the GMLC tries to map the negative reaction to a final MLP error.

This MLP error response is sent back to the LCS client. The mapping is based on the following:

- type of the responding NF node
- the HTTP status code
- cause code

When the unique combination of the following is not a supported mapping, the final MLP error code is always 6:

- type of the responding NF node
- the HTTP status code
- cause code

The supported MLP 3.2 error codes with the associated mappings are detailed in the subsequent sections.

2.5.1 MLP Error 6: POSITIONING METHOD FAILURE

Responding NF Node	HTTP Status Code	Cause Code(s)
AMF	403	UNSPECIFIED
	500	POSITIONING_FAILED



2.5.2 MLP Error 501: ROAMING ERROR

Responding NF Node	HTTP Status Code	Cause Code(s)
UDM	403	ROAMING_NOT_ALLOWED

2.5.3 MLP Error 2: UNSPECIFIED ERROR

Responding NF Node	HTTP Status Code	Cause Code(s)
UDM	403	INVALID_GUAMI
	422	UNPROCESSABLE_REQUEST
AMF	403	UNAUTHORIZED_REQUESTING_NETWORK
SCP	500	SYSTEM_FAILURE

2.5.4 MLP Error 202: POSITIONING NOT ALLOWED

Responding NF Node	HTTP Status Code	Cause Code(s)
UDM	403	ACCESS_NOT_ALLOWED
		DNN_NOT_ALLOWED REAUTHENTICATION_REQUIRED
AMF	403	POSITIONING_DENIED

2.5.5 MLP Error 4: UNKNOWN SUBSCRIBER

Responding NF Node	HTTP Status Code	Cause Code(s)
UDM	403	UNKNOWN_5GS_SUBSCRIPTION NO_PS_SUBSCRIPTION
	404	USER_NOT_FOUND
AMF	403	USER_UNKNOWN
	404	CONTEXT_NOT_FOUND



2.5.6 MLP Error 5: ABSENT SUBSCRIBER

Responding NF Node	HTTP Status Code	Cause Code(s)
UDM	404	CONTEXT_NOT_FOUND
AMF	403	DETACHED_USER
	504	UNREACHABLE_USER

2.5.7 MLP Error 1: SYSTEM FAILURE

Responding NF Node	HTTP Status Code	Cause Code(s)
AMF	504	PEER_NOT_RESPONDING
		TARGET_NF_NOT_REACHABLE
SCP	500	SCP_RELATED_PROCESSING_FAILURE



3 Operation and Maintenance

3.1 Events

The NLG supports the following events for the 5G GMLC within the HTTP2 component:

- TLS Certificate events
- NRF server availability events

Note: The All NRF servers are unavailable event can affect all traffic flows on Dual or Multi setups. The event does not affect Single setups, as it causes all traffic interfaces to close on the same NLG server where it appeared, and migrates all traffic handling to other available NLGs. If all the NLGs are unable to connect to any of the configured NRF servers, and 5GS is in use, then complete traffic outage is expected for all traffic throughout the cluster. For more information on server availability, see *Availability* in Technical Product Description, NLG.

- AMF and UDM connectivity events
- AMF and UDM communication events

For detailed information about the events, see *Event Description*, NLG.

3.2 Alarms

The NLG supports the following alarms for the 5G GMLC within the HTTP2 component:

- Network Connection to 5G Core (NRF)
- TLS Related Alarm
- Connection towards AMFs and UDMs

For detailed information about the alarms, see *Alarm Descriptions*, NLG.

3.3 Statistics

The NLG generates statistics counters for 5G GMLC positioning. The counters record the following information:

- The number of successful and unsuccessful queries towards the NRF servers



- The unsuccessful communication between the NLG node and the AMF and UDM nodes

For more information, see [Statistics Description, NLG](#).

3.4 Positioning Records

For the 5G GMLC, the NLG supports XML-based positioning records for the location requests, including additional elements for the NR network.

Note: The data types of the location data in the NR network are according to *3GPP 29.572*. These data types are different from other network access types.

For more information, see [Positioning Record File Description, NLG](#).

3.5 Configuration Management

The configuration of the 5G GMLC feature includes the following: Component Parameters on NLG components.

The 5G GMLC serves only as a client in the 5G Core network, and the interface through which it communicates is non-configurable. The route is determined by Linux routing table based on the IP address of the desired 5G core node.

For more information, see [Parameter List, NLG](#).

3.6 Feature Activation

The 5G GMLC feature can be enabled and disabled from the Administration Portal under the **Networking Connection > Mobile Core Network > AMF/UDM**. The switch is called **Access HTTP2**.

Note: If the 5G GMLC feature is disabled, but the NLG attempts to serve 5G traffic, it returns error messages to the LCS client.

3.7 Cell Data Management

In the NCGI translation case of the 5G GMLC positioning, the NLG must be aware of the geographic location of the NR cells. The NR cell data can be imported from an XML file, provisioned through the CLI or the Administration Portal.

For more information, see [NR Cell Data File and CLI Description, ENL](#).



3.8 License Handling

The following licenses are needed for the 5G GMLC in the NLG:

- To support 5G GMLC commercial positioning, the NLG GMLC Value Package (VP) license for GMLC must be installed.
- To support 5G GMLC emergency positioning, the NLG Emergency Location Base Package must be installed.

For more information on licenses, see [Technical Product Description, NLG](#).

3.9 Administration Portal

The Administration Portal provides a web-based interface to manage the 5G GMLC feature. It includes the following functions:

- Network settings for the 5G GMLC
- Data management for cell data
- Software management for the 5G GMLC-related components
 - HTTPServer
 - MLP
 - Request Monitor
 - Network Storage
 - PPRouter
 - HTTP2
- System records
 - Statistics
 - Positioning records
 - Billing
- System monitor
 - Event Log
 - Alarm Log
 - Active Alarm List



For more information, see [Administration Portal User Guide for ENL Cloud Native, NLG](#).

3.10 High Availability

The 5G GMLC feature does not support High Availability from the 5G core network side, as it acts as a client towards the NFs.

Ingress LCS client traffic is distributed by the load balancer in case of a system comprised of two or more NLG servers. For more information, see [System Administration Guide for ENL Cloud Native, NLG](#).

3.11 Billing

For the 5G GMLC feature, the billing file is amended with the AMF InstanceId (numberAMF) and the 5G network type (NR). For more information, see [Billing File Description](#).



4 Engineering Guidelines

Ericsson recommends the following engineering guidelines:

The 5G software design is based on the assumption that the product will be integrated into the network using the mTLS as mandated by the 3GPP standard. This ensures mutual authentication between the network elements, provides integrity protection for messages and encrypts the data stream, which may contain sensitive data. In this regard, there is no other control built into the ENL in case of disabled mTLS.

Disabling mTLS is only recommended after a comprehensive risk analysis, taking into account the possible technical and business impacts.

For TLS, Ericsson strongly recommends installing certificates issued by a trusted certificate authority (CA) instead of using self-signed certificates. 5G GMLC uses three separate CA files to validate nodes it communicates with from the 5G Core. One for all the NRF, one for all the UDM, and one for all the AMF servers. For information on how to install customized certificates, see the *Secure Communication* section in *System Administration Guide for ENL Cloud Native, NLG*.